

Supplementary Figure 1. Photos illustrating the depth-resolved soil sampling method described in Methods 2.2. Photos are arranged to represent the method chronologically from left to right, but each shows the sampling area under a different tree (labeled below). In the second row, a photo of Tree 04 is annotated to show the location of the sampling area for spatial context.



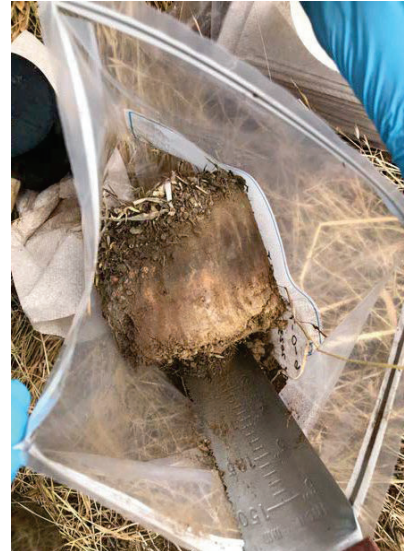
Tree 03



Tree 02



Tree 04



Tree 06



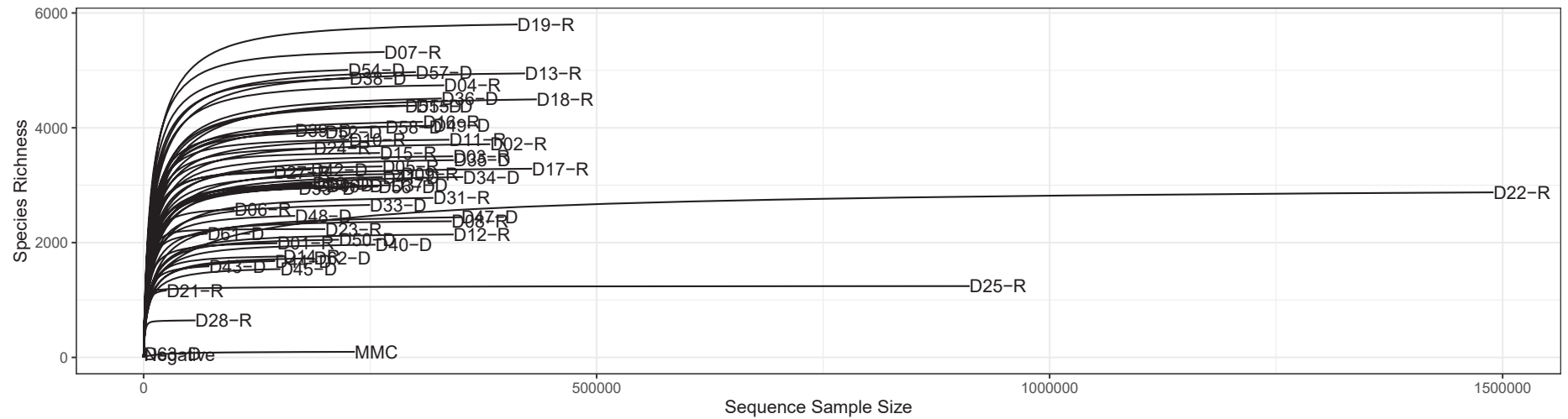
Tree 05



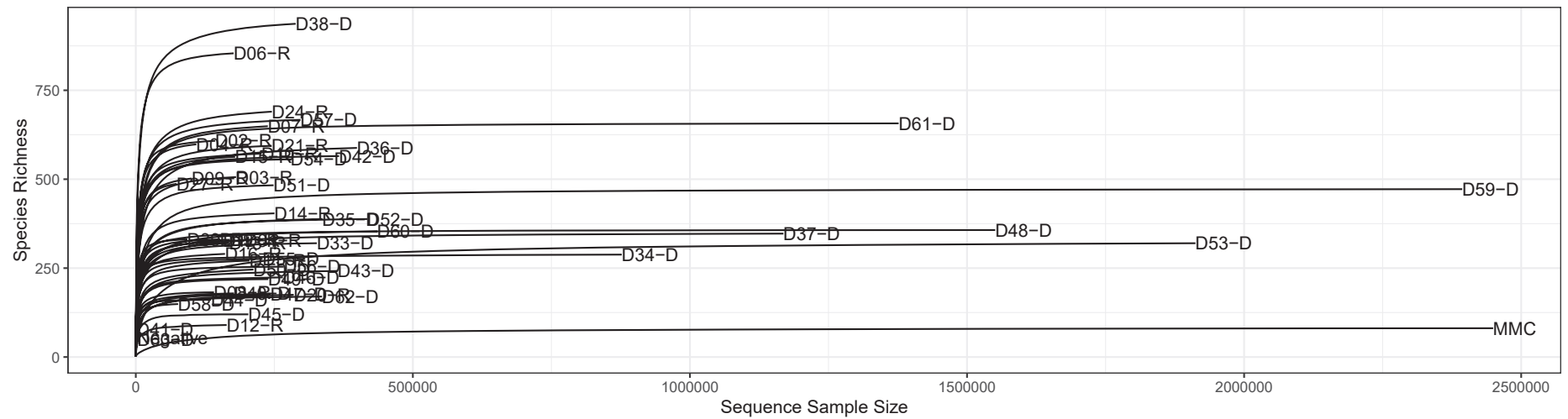
Tree 04

Supplementary Figure 2. Rarefaction curves for 16S and ITS metabarcoding data. 'Species Richness' refers to number of unique ASVs observed in a single sample after quality-control trimming (removed 3 reads from all 16S ASVs, 8 reads from all ITS ASVs).

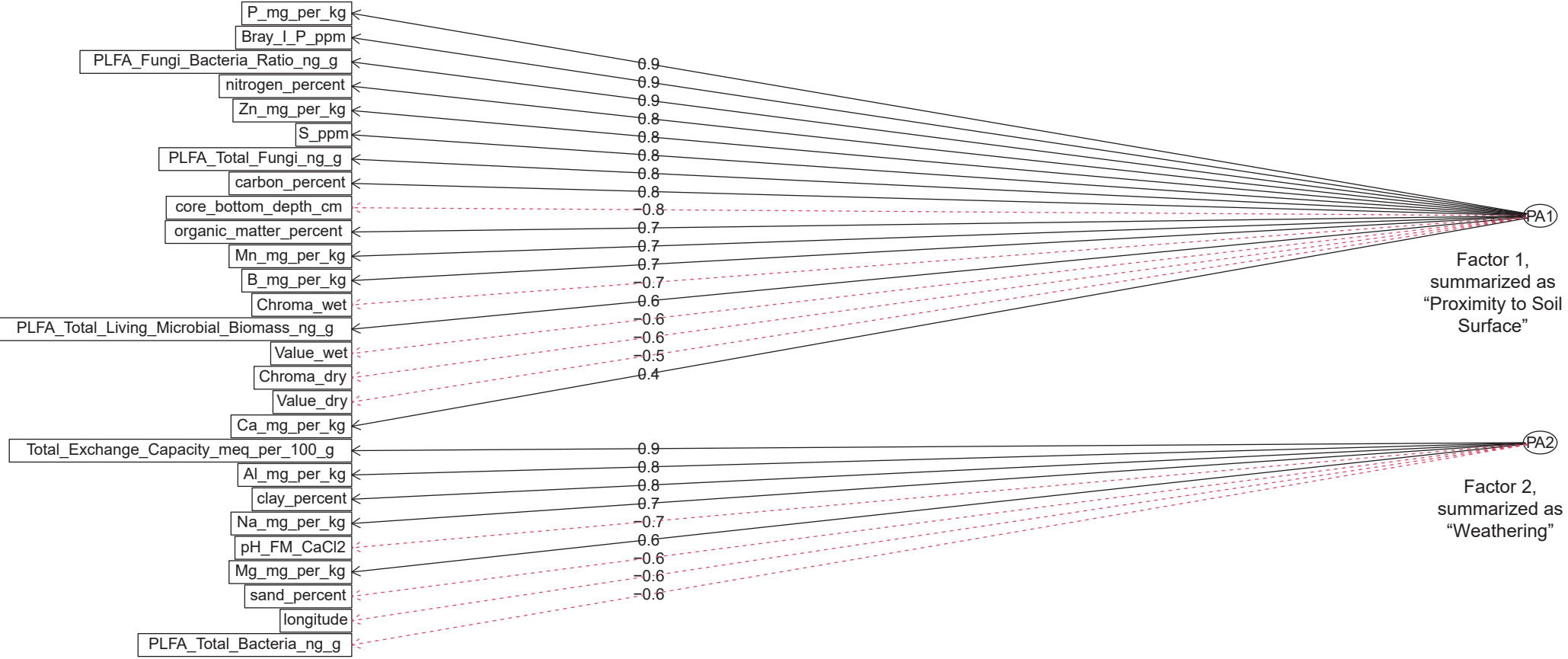
16S



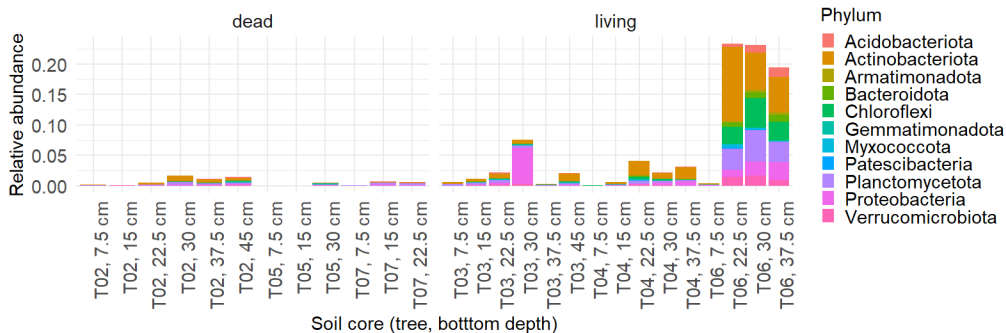
ITS

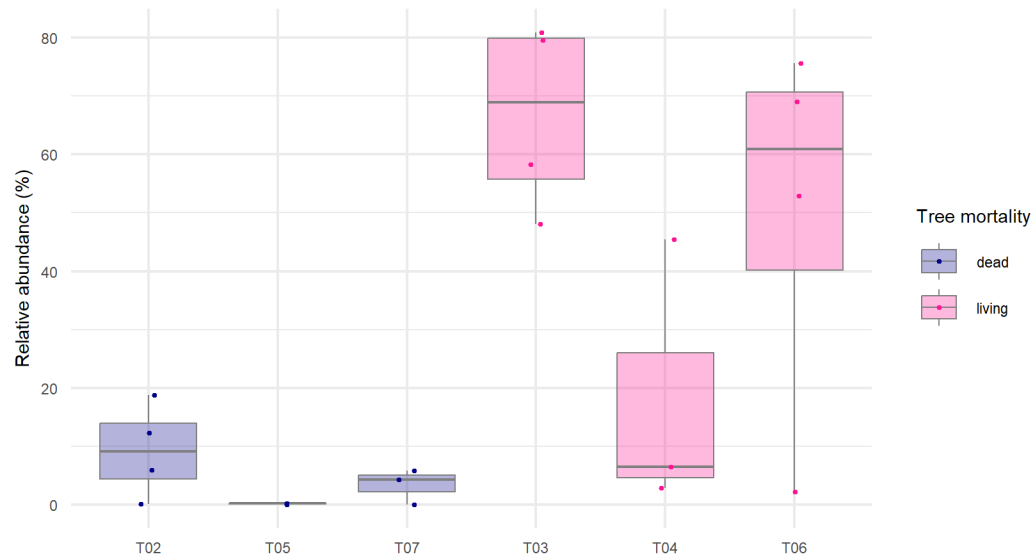


Supplementary Figure 3. Correlations between soil edaphic variables and the calculated factors displayed in Figure 1. We estimate that F1 (shown as PA1 below) corresponds to soil depth, while F2 (shown as PA2 below) corresponds to increased soil development (weathering). Factors were computed in an exploratory factor analysis using the psych package in R (Supplementary Table 3).



Supplementary Figure 5. Relative abundance of 'ectomycorrhizal-promoted' ASVs within whole-sample bacterial communities as measured by 16S rRNA metabarcoding. 'Ectomycorrhizal-promoted' here refers to ASVs whose crude abundance in a soil sample was positively correlated with the imputed biomass of ectomycorrhizal fungi. The y-axis represents the proportion (0–1) of the RNA-based bacterial community comprised by these ASVs within each discrete soil sample. On the X-axis, soil sample names denote both tree identity and soil depth in cm, and are clustered by tree identity. ASVs are colored according to their phylum.

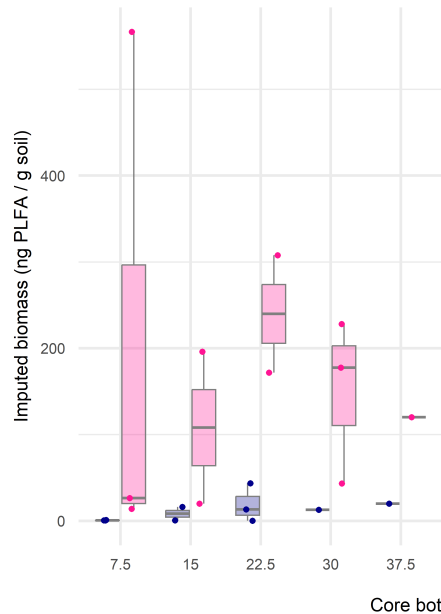




Supplementary Figure 6. Box and whisker plot showing the relative abundance of ectomycorrhizal fungi underneath each tree, including data from all soil depths for which fungal community (as measured by 16S rRNA metabarcoding) data was available. Relative abundance was calculated as proportion of the fungal community assigned to the ectomycorrhizal primary lifestyle * 100. Box boundaries are the first and third quartiles. Sample size is variable between boxes due to complications with RNA extraction and amplification for metabarcoding; from left to right, $n = 4, 5, 3, 4, 4, 3$.

Supplementary Figure 4. Box and whisker plots showing the distribution of imputed biomass (A) and relative abundance (B) of ectomycorrhizal fungi across discrete soil depths. Imputed biomass was calculated as: the proportion of the fungal community (as measured by 16S rRNA metabarcoding) assigned to the ectomycorrhizal primary lifestyle * total living fungal biomass as measured by PLFA. Relative abundance was calculated as the proportion of the fungal community assigned to the ectomycorrhizal primary lifestyle * 100. Box boundaries are the first and third quartiles. Individual data points are overlaid; sample size is variable between boxes due to complications with RNA extraction and amplification for metabarcoding.

A



B

